

POWERFACTORY

PowerFactory 2024

Technical Reference

Fault Current Limiter

ElmFcl

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

This document describes the models for the fault current limiter *ElmFcl*. The model is implemented to represent a resistive type of a superconducting fault current limiter (SFCL).

2 Load flow calculation

For the load flow calculation the fault current limiter is represented with a series resistance with the steady state value *R*.

3 Harmonic analysis

For the power quality and harmonic analysis, the model is represented with a series resistance (steady state value R). A possibility is provided to define a frequency dependency on the resistance.

4 Short-circuit calculation

Generally for the short circuit calculation, the fault current limiter is represented with a series resistance with the steady state value *R*. For the complete method a more advanced model can be selected.

4.1 Complete method

For the complete method, the fault current limiter is also represented with a series resistance, but two different models are available depending if the *Consider current threshold for maximum resistance* option is disabled or enabled:

- If the option is disabled, the steady state resistance R is used as resistance.
- If the option is enabled, the fault current limiter can trigger between the superconducting state (using the steady state resistance R) and the state with maximum resistance (using $Rmax$) automatically.

  The short-circuit calculation is carried out as follows:
  1. Calculate the subtransient short-circuit current using the steady state resistance;
  2. Evaluate the subtransient short-circuit current against the current threshold $IthreshSHC$;
  3. If the subtransient short-circuit current is equal or greater then the current threshold, the subtransient short-circuit is re-calculated using the maximum resistance.
  4. The state determined for the subtransient calculation is retained for the transient calculation.

  For unbalanced short-circuit calculations, if one phase exceeds the threshold, the state of all phases is triggered.

5 RMS simulation

The model used for the RMS simulation is based on the model used for the EMT simulation (see Section 6). The following differences exist:

- The threshold current used is the rms current (calculated from the peak value as $Ithresh/\sqrt{2}$),
- For the built-in triggering, the slope of the current is not used.

6 EMT simulation

In the EMT simulation, the model is implemented to represent a resistive type of a superconducting fault current limiter (SFCL). It is a simplified model that uses an exponential function and two linear functions. The amplitude and length of the characteristic is defined by the input parameters. The triggering of the characteristic can be done using a built-in triggering or by external signals. As an alternative, the characteristic can be implemented by an external dynamic model using resistance input signals.

The model characteristic with predefined resistance values and duration is shown in Figure 6.1 where:

- R is the steady state (superconducting) resistance value,
- $Rmax$ is the maximum resistance value,
- $Rrec$ is the resistance value between the recovery periods,
- $t_trigger$ is the triggering time when the model switches from superconducting into quenching state,
- tq is the duration of the quenching period (state),
- $t1$ is the duration of the first recovery period (state),
- $t2$ is the duration of the second recovery period (state).

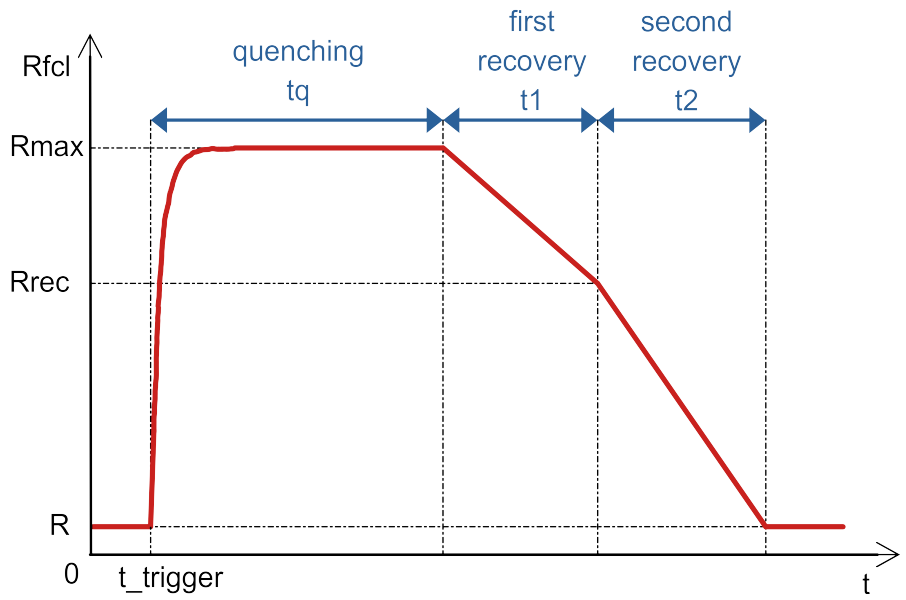


Figure 6.1: Resistant SFCL characteristic

The nonlinear resistance in ohm of each phase is modelled by using four different states:

- Superconducting state
  In this state the element is represented by the steady state resistance:

$$Rfcl = R \tag{1}$$

- Quenching state
  In this state the element is represented by an exponential function:

$$Rfcl = R + (Rmax - R) \cdot \sqrt{1 - e^{\left(\frac{-(t - t_trigger)}{tau}\right)}} \tag{2}$$

  where tau is the time constant of the exponential function and t is the simulation time,

- First recovery state
  In this state the element is represented by a linearly changing resistance:

$$Rfcl = \frac{Rrec - Rq}{t1} \cdot (t - (t_trigger + tq)) + Rq \tag{3}$$

  where Rq is the resistance value at the end of the quenching process.

- Second recovery state
  In this state the element is also represented by a linearly changing resistance:

$$Rfcl = \frac{R - Rrec}{t2} \cdot (t - (t_trigger + tq + t1)) + Rrec \tag{4}$$

6.1 Built-in triggering unit

The built-in triggering works separately for all phases. The following state changes are possible:

- From superconducting to quenching state

  During the superconducting state, the absolute value of the peak value of the phase current is monitored and compared to the threshold value $Ithresh$. In addition, the slope of the peak current is also compared to the threshold value of $Islopeth$ (this condition can be disabled by setting the value to zero).

  When the conditions are met, the quenching state is triggered.

- From quenching to first recovery state

  After a time period of tq (*Duration of quenching period*), the state is automatically changed to the first recovery state.

- First recovery to second recovery state
  After a time period of $t1$ (*Duration of first recovery period*), the state is automatically changed to the second recovery state.

- From second recovery to superconducting state
  After a time period of $t2$ (*Duration of second recovery period*), the state is automatically changed to the superconducting state.

6.2 External triggering

If the internal *Built-In triggering unit* is not used, alternatively, the triggering per phase can be executed by the external signals *trigger_A*, *trigger_B* and *trigger_C* by using either a parameter event (*EvtParam*) defined for the EMT simulation, or via an external controller.

6.3 Resistance inputs

A different resistance characteristic can be implemented by using the external signals *Rin_A*, *Rin_B* and *Rin_C* with which the resistance value per phase can be modified externally.

7 Input signals of the RMS and EMT models

Input signal	Symbol	Description	Unit
trigger_A		Trigger phase a	
trigger_B		Trigger phase b	
trigger_C		Trigger phase c	
Rin_A		Resistance phase a	Ohm
Rin_B		Resistance phase b	Ohm
Rin_C		Resistance phase c	Ohm

Table 7.1: Input signals of the unbalanced RMS and EMT models

Input signal	Symbol	Description	Unit
trigger		Trigger	
Rin		Resistance	Ohm

Table 7.2: Input signals of the balanced RMS, single-phase and dc elements

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